Business Process Automation

with Power Automate and Copilot Studio Agents

WSQ Course Code: TGS-2022017524 · 2-Day Hands-On Training · Modules 1–4

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer/administrator displays the digital attendance QR code from the SSG portal.
- Scan the QR code with your mobile phone camera and submit your attendance.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

About the Trainer

?

Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

YOUR TRAINER

About the Trainer

AA

Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte. Ltd.

ROLE

Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

CERTIFICATION

Microsoft Power Platform, AI and data analytics — 20+ years of training experience.

DELIVERS

WSQ courses on AI, business automation, data science and software engineering.

FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

ICE-BREAKER

Let's Know Each Other

- Your name and organisation / role.
- Your experience with Power Automate, Copilot Studio or other automation tools (if any).
- One repetitive business process you would love to automate after this course.

HOUSEKEEPING

Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively — no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. 75% attendance is required.

COURSE OUTCOMES

What You'll Learn

Power Automate flows → add AI → **Copilot Studio agent** → publish → **Teams + website channels** → take action → **Agent tools and prompts**

Day 1 — automate

Build reliable trigger → action flows without an AI agent.

Day 2 — converse

Create a grounded agent that understands and confirms user intent.

Integrate — act

Let the agent call deterministic or prompt-based Power Automate tools.

Your 2-Day Journey

1 Form email
Confirm the submitted user

2 Excel log
Add an audit row

3 Event branch
Route Yes and No

4 Leave approval
Manager decides

5 IT agent
FAQ + Teams

6 HR agent
SharePoint + Teams

7 Agent routing
IT or HR reply

8 HTTP form
Website webhook

9 Finance chat
Grounded agent

10 Trading advisor
Candles + news

Day 1 builds Forms, flows and support agents. Day 2 adds webhooks and the Finance Advisor capstone.

How This Course Works

Concept, then demo, then hands-on — for every topic

Built entirely in Microsoft 365 and the Power Platform

No coding required — low-code, drag-and-drop tools

Save your flows and agents to reuse in your own role

- Daily recap and open Q&A to lock in what you learned

SCHEDULE

Lesson Plan — 2 Days, 9:00am–6:00pm

D1 Day 1 — Build and route

9:00–10:40 Concepts + flow types
10:55–3:15 Labs 1–4: Forms, Excel, event branch, approval
3:30–5:50 Agent concepts + Labs 5–7
5:50–6:00 Evidence check and recap

D2 Day 2 — Connect and advise

9:00–10:05 HTTP requests and webhooks
10:05–12:15 Lab 8 + Finance Agent foundation
1:15–4:00 Labs 9–10: web chat and trading advisor
4:00–6:00 Written + practical assessment

BEFORE THE ASSESSMENT

Briefing for Assessment

- Place phones and other materials under the table or on the floor.
- No photos or recording of assessment scripts.
- No discussion during the assessment.
- Use a black/blue pen for hard-copy assessments.
- No liquid paper / correction tape.
- Scripts are collected when time is up.

FINAL ASSESSMENT

Assessment & Funding

1 Written

1 hour · open book
Modules 1–4 and the lab concepts

2 Practical

1 hour · open book
Build and verify a working flow and agent

3 Evidence

Submit on the LMS
Competent result + attendance requirement

ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM survey — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit WA (SAQ) then PP — open book
4. Submit your answers on the LMS
5. Sign the Assessment Summary Record

COURSE REPOSITORY

Access the Hands-On Labs

1

10 labs

Lab 0 setup plus Labs 1–10 and four concept modules.

2

Ready assets

Excel tables, searchable PDFs, HTML pages and JSON schemas.

3

Use the guide

Open each index.md or the compiled Learner Guide and work in order.

Day 1 — Forms, Flows and Business Agents

DAY 1

Choose the trigger → Capture form data → Route and approve → Ground agents → Email the result

Build one connected automation story, then add specialised agents.

A Business Workflow Has a Clear Start and Result

Business event → Trigger → Actions → Output → Evidence

Teaching point

Name the start event, required work, expected result and proof before opening the designer.

CORE PATTERN

Trigger → Actions → Output

Exactly one trigger → Get or validate data → Perform work → Notify or return

Teaching point

Every lab uses the same skeleton; only the trigger, actions and business rules change.

Three Cloud Flow Types

1 Instant flow

A person deliberately selects Run, presses a button or invokes the flow from an app.

Use when the user controls the exact start time.

2 Scheduled flow

A Recurrence trigger reaches a defined time.

Use for daily, weekly or other timetable-driven work.

3 Automated flow

A connected-service event occurs, such as a new form response.

Use when the process must react immediately.

DECISION RULE

Choose the Flow Type by Asking One Question

A person? → **A clock?** → **A new event?**

Teaching point

Person = instant. Clock = scheduled. New record, form, file or message = automated.

FLOW COMPARISON

Same Actions, Different Triggers

1 **Instant**

Manual trigger → send email → add Excel row

2 **Scheduled**

Recurrence → read pending items → send reminder

3 **Automated**

Form response → get details → email / log / approve

AUTOMATED CLOUD FLOWS

The Microsoft Forms Trigger Pattern

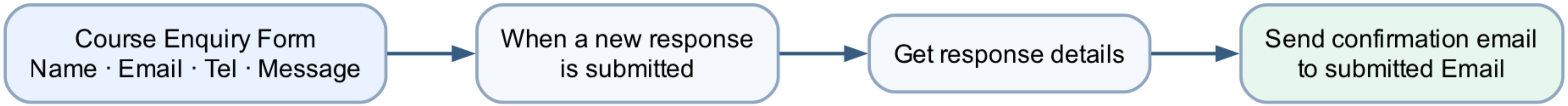
Form submitted → Response Id → Get response details → Use answers

Teaching point

The trigger identifies the response; Get response details retrieves the actual answers.

SHARED WORKFLOW VISUAL

Lab 1 — Form to Email Confirmation



Shared workflow visual — follow the arrows; decision branches are labelled.

REQUIRED FIELDS

Lab 1 — Design the Form

1 Identity

Name
Email

2 Contact

Tel

3 Enquiry

Message (long answer)

FORM ANSWERS BECOME TOKENS

Lab 1 — Map Dynamic Content

Name token → Email token → Message token → Personalised email

Teaching point

Insert tokens from Get response details. Typed labels are not dynamic content.

EVIDENCE

Lab 1 — Test the Real Outcome

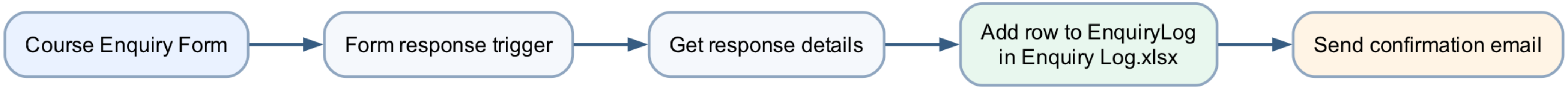
Submit once → Inspect run → Check recipient → Confirm one email

Teaching point

One form submission should produce one successful run and exactly one message.

SHARED WORKFLOW VISUAL

Lab 2 — Log the Enquiry and Send Email



Shared workflow visual — follow the arrows; decision branches are labelled.

NAMED EXCEL TABLE

Lab 2 — Enquiry Log.xlsx

Timestamp → Name + contact → Message → Status + source

Teaching point

Excel Online actions require a named table: EnquiryLog.

RELIABILITY

Lab 2 — Action Order

Get details → Add row → Send email → Verify workbook

Teaching point

Put logging before email when confirmation should be sent only after a successful record.

COMMON CHECKS

Lab 2 — Troubleshooting

1 Workbook

Store in OneDrive for Business or SharePoint.

2 Table

Select EnquiryLog; loose cells are not a table.

3 Lock

Close desktop Excel if the connector reports a file lock.

SHARED WORKFLOW VISUAL

Lab 3 — Event Registration Branching

Shared workflow visual — follow the arrows; decision branches are labelled.

Lab 3 — Yes Branch

Joining = Yes → Add EventLog row → Email admin → Participant details

Teaching point

The administrator receives the person's name, email and telephone number.

PARTICIPANT DECLINES

Lab 3 — No Branch

Joining = No → Add EventLog row → Email user → Invite next time

Teaching point

The user receives thanks and a respectful next-event message.

TEST MATRIX

Lab 3 — Branch Evidence

1 **Yes test**

Correct row + administrator email

2 **No test**

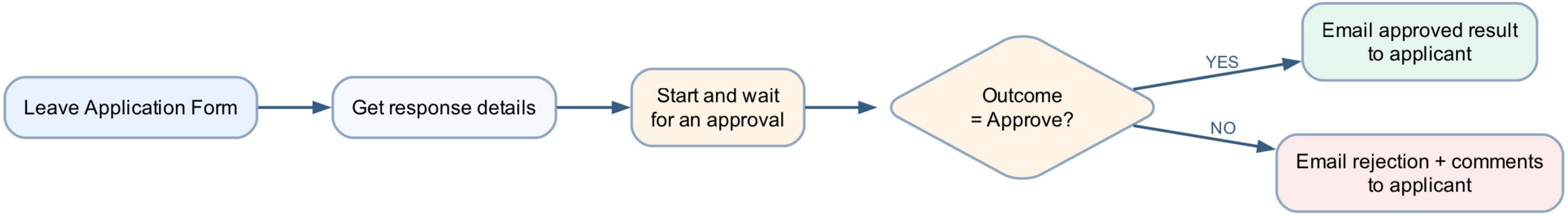
Correct row + user email

3 **Audit**

JoiningEvent and NotificationSent explain the outcome

SHARED WORKFLOW VISUAL

Lab 4 — Leave Application Approval



Shared workflow visual — follow the arrows; decision branches are labelled.

REQUIRED INFORMATION

Lab 4 — Leave Form

1 Employee

Name
Email

2 Dates

Leave from date
Leave end date

3 Request

Leave type
Reason for leave

Lab 4 — Approval Outcome

Start and wait → Outcome → Approve branch → Reject branch

Teaching point

Compare Outcome to Approve and include manager comments in the applicant email.

HANDLE PEOPLE DATA CAREFULLY

Lab 4 — Governance

1 Access

Use only authorised approvers.

2 Privacy

Limit medical and personal information.

3 Evidence

Retain decision, comments and run history.

Copilot Agent — Six Building Blocks

1 Knowledge base

Approved files, websites or SharePoint sources used to ground answers.

2 Skills

Capabilities realised through topics, tools and agent flows.

3 Tools

Connected functions that read data or create authorised outcomes.

PART 2

Copilot Agent — Six Building Blocks

1 Memory

Conversation context and variables; persistence depends on enabled features and governance.

2 Model

The generative model interpreting requests and composing responses.

3 System message

Instructions defining role, scope, tone, safety and escalation.

Knowledge Answers; Tools Act

User asks → Instructions set scope → Knowledge grounds → Tool acts → Agent replies

Teaching point

Use static knowledge for approved explanations and tools for live, authorised actions.

BEFORE PUBLISHING

Agent Safety Test Set

1 Expected

Questions the agent should answer.

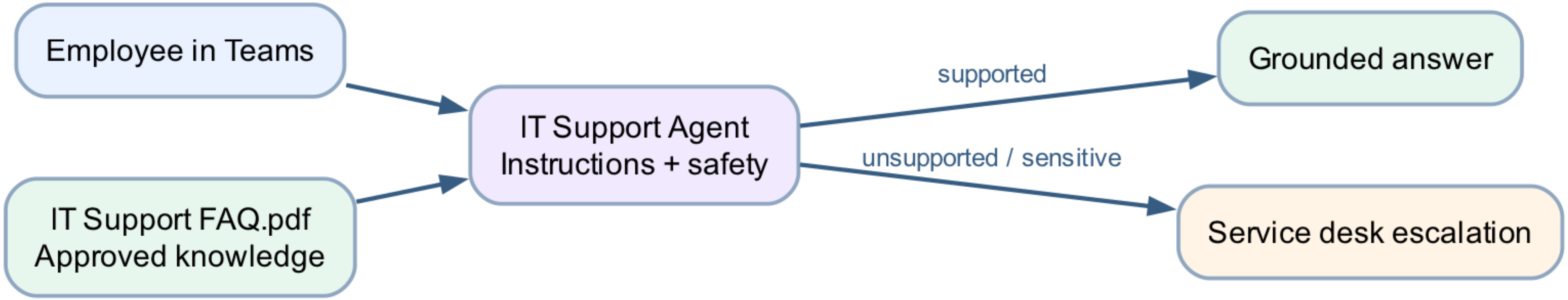
2 Unsupported

Missing or out-of-scope information.

3 Unsafe

Credentials, personal data or prohibited decisions.

Lab 5 — IT Support Agent



Shared workflow visual — follow the arrows; decision branches are labelled.

Lab 5 — Test and Deploy

1 Positive

Password reset, MFA, VPN and device loss.

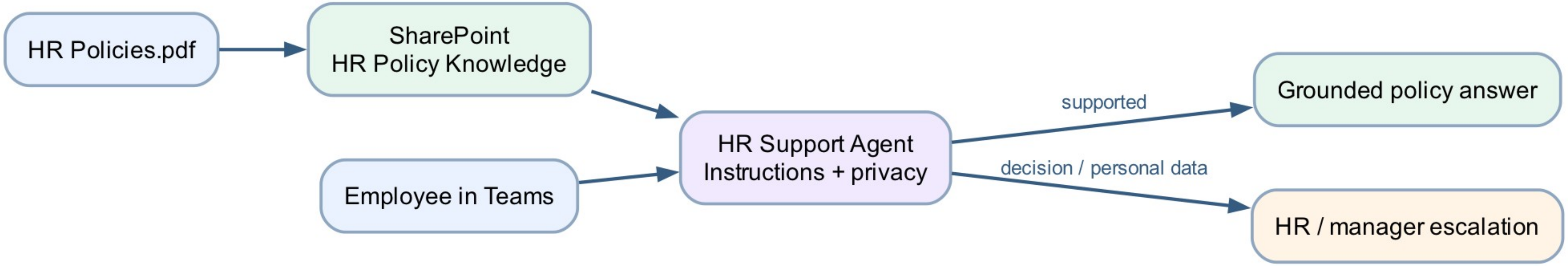
2 Negative

Credential request and unrelated HR question.

3 Channel

Publish latest version and verify in Teams.

Lab 6 — HR Support Agent



Shared workflow visual — follow the arrows; decision branches are labelled.

INFORMATION, NOT DECISIONS

Lab 6 — HR Boundaries

1 Explain

Approved leave, expenses and workplace policy.

2 Protect

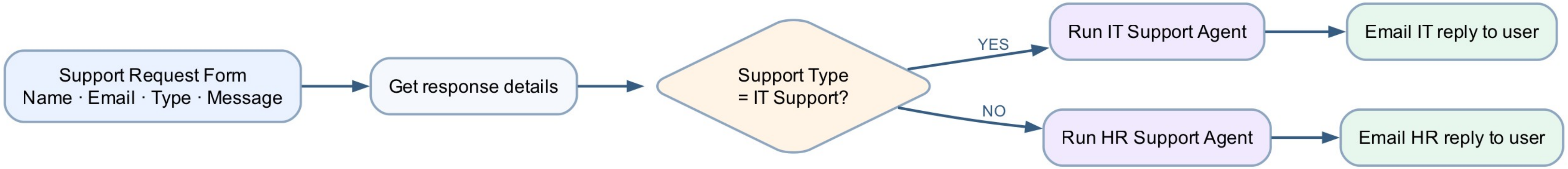
Never expose another employee's records.

3 Escalate

HR and managers make final decisions.

SHARED WORKFLOW VISUAL

Lab 7 — Route Requests to Specialised Agents



Shared workflow visual — follow the arrows; decision branches are labelled.

ONE CONNECTED PROGRESSION

Day 1 Recap

1

Flows

Forms → email → Excel → condition → approval

2

Agents

IT FAQ and HR SharePoint knowledge

3

Integration

Form → specialised agent → user email

COURSE JOURNEY

Day 2 — HTTP, Webhooks and Agent Websites

DAY 2

HTTP request → Webhook URL → Finance chat → Market tools → Trading website

Expose flows safely to websites and finish with a tool-using Finance Advisor Agent.

An HTTP Request Has Four Main Parts

Method → **URL** → **Headers** → **Body**

Teaching point

The lab websites use POST to send a JSON body to the Power Automate endpoint.

INTENT

Common HTTP Methods

1 GET

Read data

2 POST

Submit data or start work

3 PUT · PATCH · DELETE

Update or remove data

A Webhook Is an Event Endpoint

Website → POST JSON → HTTP trigger → Flow actions → JSON Response

Teaching point

Power Automate generates the URL after a valid trigger-and-action flow saves.

Webhook Safety

1 URL

Treat an anonymous webhook URL like a secret.

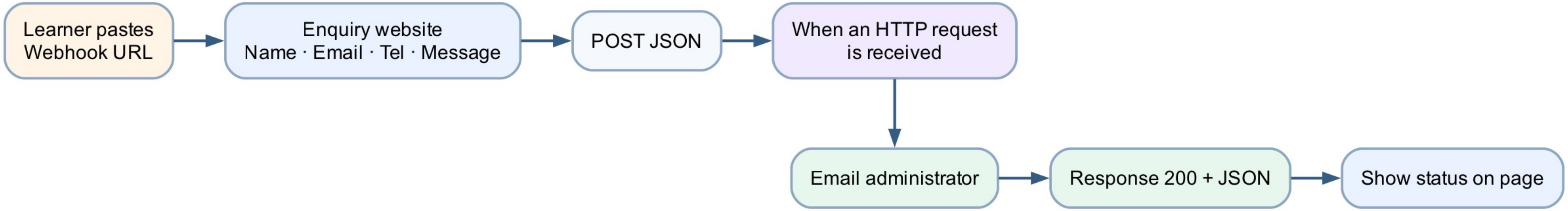
2 Input

Validate, minimise and never include credentials.

3 Output

Return only the data the website needs.

Lab 8 — Website HTTP Enquiry



Shared workflow visual — follow the arrows; decision branches are labelled.

JSON SCHEMA

Lab 8 — Request Contract

name → **email** → **tel** → **message**

Teaching point

Generate the trigger schema from a sample payload before mapping fields.

Lab 8 — Complete the Flow Before Saving

HTTP trigger → Send email action → Response action → Save → Copy URL

Teaching point

A trigger alone is incomplete; add at least one action and resolve validation errors.

END-TO-END EVIDENCE

Lab 8 — Browser Test

1 Connect

Paste and save the current webhook URL.

2 Submit

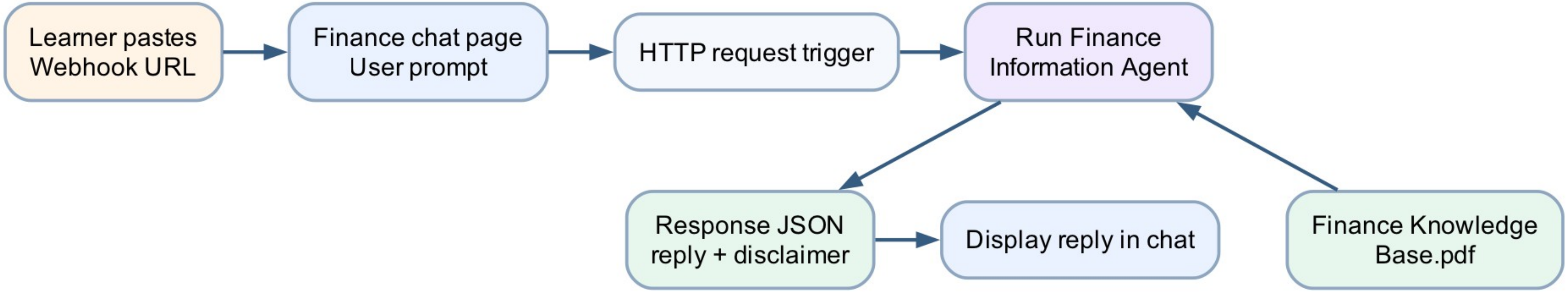
Enter realistic classroom data once.

3 Verify

Page status, run history and administrator email.

SHARED WORKFLOW VISUAL

Lab 9 — Finance Agent Web Chat



Shared workflow visual — follow the arrows; decision branches are labelled.

EDUCATIONAL USE

Lab 9 — Finance Knowledge Boundary

1 Explain

Orders, candles, timeframes, volatility and news.

2 Qualify

Separate facts from interpretation and uncertainty.

3 Refuse

No guarantees or personalised buy/sell instructions.

Lab 9 — HTTP-to-Agent Flow

prompt → Run Finance Agent → agent response → Response JSON

Teaching point

Insert the actual agent-response output token into the JSON reply.

Lab 9 — Test Set

1 **Concept**

Market versus limit order.

2 **Uncertainty**

Question not covered by knowledge.

3 **Boundary**

Request for guaranteed returns.

LEARNER-ENTERED WEBHOOK

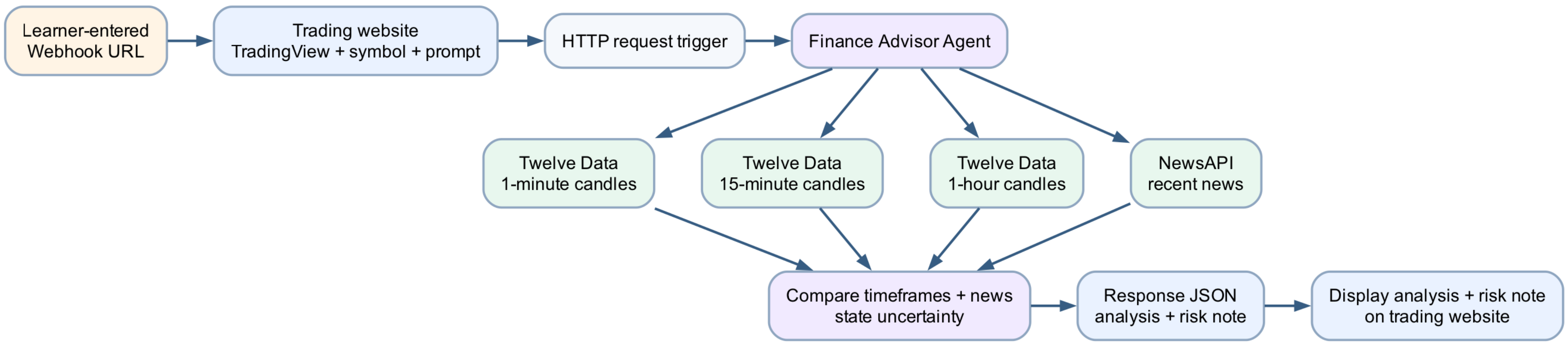
Lab 9 — Connect the Chat Page

Save URL locally → **Send prompt** → **Display reply** → **Show disclaimer**

Teaching point

One message should create one flow run; do not embed tenant URLs in source.

Lab 10 — AI Trading Advisor Website



Shared workflow visual — follow the arrows; decision branches are labelled.

Lab 10 — Four Authorised Tools

1 Twelve Data

1-minute candles
15-minute candles
1-hour candles

2 NewsAPI

Recent relevant company or symbol news

3 Guardrails

Validate symbols, limits, errors and credentials

DO NOT FORCE AGREEMENT

Lab 10 — Multi-Timeframe Reasoning

1-minute → 15-minute → 1-hour → Compare direction → Explain conflict

Teaching point

Short intervals contain more noise; a sound response preserves conflicting evidence.

RECENT INFORMATION

Lab 10 — Add News Context

Query news → Check time + source → Relate cautiously → State gaps

Teaching point

A headline can be delayed, incomplete or already reflected in price.

NO KEYS IN THE BROWSER

Lab 10 — Secure API Design

1 Store

Protected connections, environment variables or custom connector.

2 Limit

Small result sets and provider rate limits.

3 Hide

Never expose keys in HTML, prompts, logs or screenshots.

MODEL BOUNDARIES

Lab 10 — Finance Advisor Instructions

Use all tools → Separate data and inference → Describe uncertainty → End with risk note

Teaching point

The agent provides education; it never promises returns or tells a person what to buy.

Lab 10 — HTTP Request Contract

symbol → prompt → Validate + normalise → Run agent → Return JSON

Teaching point

Reject invalid symbols and return a clear error instead of fabricated analysis.

Lab 10 — TradingView Website

Enter symbol → Update chart → Paste webhook → Ask agent → Render response

Teaching point

TradingView visualisation and agent analysis are related but independent components.

EXPECTED FAILURES MATTER

Lab 10 — Test Matrix

1. **Valid**

   All data tools succeed and output is qualified.

2. **Partial**

   Missing news or conflicting timeframes are disclosed.

3. **Invalid**

   Bad symbol or advice request gets a safe response.

OBSERVE → REASON → ACT → RESPOND

End-to-End Agentic Architecture

Website user → **Power Automate** → **Copilot agent** → **Data tools** → **Browser response**

Teaching point

Each boundary has a distinct responsibility and an auditable input/output.

BEFORE PRODUCTION

Operational Readiness Review

1 Security

Authentication, secrets and least privilege.

2 Reliability

Timeouts, retries, rate limits and fallbacks.

3 Governance

Approved sources, disclaimers, logs and owners.

Day 2 Recap

1 Connect

HTTP request + learner-entered webhook URL

2 Ground

Finance knowledge + live authorised tools

3 Respond

Structured JSON + uncertainty + risk reminder

What You Can Automate Next

Onboarding: new-hire checklists and welcome emails

Reporting: scheduled data pulls and summaries

Helpdesk: triage and route incoming requests

Document handling: filing, approvals and reminders

- Anywhere people copy data or chase approvals by hand

Common Pitfalls to Avoid

Automating a broken process — fix the process first

Trying to build everything in one giant flow

Skipping tests, then surprises in production

No error handling — failures go unnoticed

- Vague agent prompts that return inconsistent data

Common Errors & Quick Fixes

“Unauthorized” sending email — reconnect Outlook with a mailbox-enabled account

Approval ‘valid users’ error — pick the approver from the dropdown (a real user in your tenant)

Date shows literal utcNow() — enter it via the fx / Expression editor so it becomes a token

Excel cell shows ######## — the column is just too narrow; auto-fit it

- Unwanted ‘For each’, or agent can’t see its flow — use single-value fields; same environment in both tools

Governance & Security Basics

Use connections and permissions you're authorised to use

Keep sensitive data out of emails and logs where possible

Follow your organisation's data and approval policies

Document what each flow does and who owns it

- Review and retire flows that are no longer needed

Rolling This Out at Work

Pick one painful, repetitive task to automate first

Prove the value, then share the win with your team

Build a small library of reusable flows and agents

Involve the people who do the work in the design

- Iterate — improve flows as needs change

Course Recap — The Big Picture

1 — Automate

Instant
Scheduled
Automated
Approval
HTTP

2 — Add the agent

Instructions
RAG knowledge
Teams + website
Confirmed inputs

3 — Let it act

Agent flow
Prompt flow
Guardrails
Returned result

Key Takeaways

Every workflow is Trigger → Actions → Output

Power Automate does the work; Copilot Studio adds the intelligence

Structured agent outputs make flows reliable

Orchestration turns steps into complete processes

- Start small, test, and improve — then scale

Resources & Where to Go Next

make.powerautomate.com — build and manage your flows

copilotstudio.microsoft.com — create and publish agents

Microsoft Learn — free guided learning paths

Templates gallery — start from pre-built flows

- Your community — share flows and patterns with colleagues

COURSE PORTAL

Courseware & Assessment on the LMS

- Download the slides and the Learner Guide from the LMS for the open-book assessment.
- Portal: https://lms-tms.tertiaryinfotech.com/
- Submit your Written Assessment and Practical Performance answers on the LMS.
- Complete the TRAQOM survey via the QR code on the LMS.

Assessment

- Written Assessment (SAQ) — 1 hour. Practical Performance (PP) — 1 hour.
- Open book: slides, Learner Guide and approved materials only.
- Remember to take the Assessment digital attendance (TRAQOM).
- Submit your completed answers on the LMS at https://lms-tms.tertiaryinfotech.com/.

ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM survey — scan the QR code on the LMS
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Thank You

Now go automate something — questions welcome